

Mohamed Ahmed

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Education

University of Minnesota

Expected May 2027

Bachelor of Science in Electrical Engineering

Relevant Coursework: Wireless Hardware System Design, Antenna Design, Analog Electronics, Embedded Systems, Control Systems, Power Electronics

Experience

Michels Power

May 2026 – Aug 2026

Project Management Engineering Intern

- Supported electric utility projects through field verification, GIS records, engineering drawings, and technical documentation.
- Reviewed engineering drawings and as-built records for transmission and distribution projects.
- Developed a technical database tracking **1,200+ utility as-built records** for project documentation.

Club Experience

STRIVE Medical

September 2025 – Present

Electrical / Hardware Team Lead

- Lead electrical development for a **medical robotics platform** integrating sensing, embedded electronics, wireless interfaces, and power.
- Designed a **4-layer PCBA** supporting sensor interfaces, embedded controllers, power distribution, and motor-drive hardware.
- Integrated **ESP32, Teensy 4.1, IMUs, encoders, and BLE-connected embedded hardware** across robotic subsystems.
- Performed PCB bring-up and hardware/software debugging using **oscilloscopes, logic analyzers, and multimeters**.
- Led electrical development to **unanimous 1st Place** at the 2026 UMN MVP Challenge.

Projects

Wireless Hardware System Design | RF Systems, Circuit Analysis, MATLAB

- Analyzed **RF receiver and transmitter architectures** and the role of LNAs, mixers, power amplifiers, and frequency-generation blocks in wireless transceivers.
- Evaluated RF system performance using **gain, noise, linearity, bandwidth, sensitivity, and dynamic range**.
- Applied **S-parameters, transmission-line theory, impedance matching, and Smith charts** to RF circuit analysis.
- Analyzed **noise figure, cascaded noise, harmonic distortion, intermodulation, and gain compression** in RF systems.

3.5 GHz Helical Antenna | RF Design, HFSS, Electromagnetics

- Designed and simulated an **axial-mode helical antenna centered near 3.5 GHz** using ANSYS HFSS for circularly polarized wireless operation.
- Performed parametric sweeps of **helix pitch and number of turns** to optimize impedance matching, gain, bandwidth, and radiation behavior.
- Analyzed **S11, VSWR, input impedance, radiation patterns, directivity, and realized gain** across approximately 2.5–4.0 GHz.
- Achieved simulated peak gain of approximately **14 dBi** in optimized configurations while evaluating antenna geometry and matching tradeoffs.
- Compared return-loss and far-field simulation results to evaluate antenna performance across candidate geometries.

Skills

RF Systems: Receiver/Transmitter Architectures, Gain, Noise Figure, Linearity, Sensitivity, Dynamic Range

RF Circuits: LNAs, Mixers, RF Amplifiers, S-Parameters, Impedance Matching, Two-Port Networks, Stability

Microwave & Antennas: Transmission Lines, Smith Chart, S11, VSWR, Radiation Patterns, Gain, Directivity

Analog & Mixed-Signal: Analog Circuits, Noise Analysis, ADCs, Signal Conditioning, Analog Filtering

PCB & Hardware: PCB Design, 4-Layer PCBs, Grounding, Sensor Interfaces, PCB Bring-Up, Hardware Debugging

Tools: ANSYS HFSS, MATLAB, LTspice, Altium Designer, KiCad, Oscilloscope, Logic Analyzer, Git